INTRUSION DETECTION SYSTEM USING SYNTHETIC NETWORK DATA TRAFFIC

A PROJECT REPORT

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in partial fulfillment for the award of the degree

of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE & ENGINEERING

of

FACULTY OF ENGINEERING AND TECHNOLOGY

SRM INSTITUTE OF SCIENCE & TECHNOLOGY, NCR CAMPUS

April 2026

SRM INSTITUTE OF SCIENCE & TECHNOLOGY

(Under Section 3 of UGC Act, 1956)

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Certified that this project report titled "**INTRUSION DETECTION SYSTEM USING SYNTHETIC NETWORK DATA TRAFFIC**" is the bonafide work of " **Sayyad Saif Ali [Reg No: RA2311030030013], Dhruv Mittal [Reg No: RA2311030030022], Saksham Jaiswal [Reg No. RA2311030030041], Harshit Srivastava [RA2311030030004]**", who carried out the project work under my supervision. Certified further, that to the best of my knowledge the work reported herein does not form any other project report on the basis of which a degree or award was conferred.

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ABSTRACT

In the rapidly evolving digital landscape, cybersecurity has become a cornerstone of network reliability and data protection. With the growth of online services, cloud computing, and connected devices, organizations face constant threats from malicious intrusions that can compromise sensitive information, disrupt operations, and cause financial losses. Traditional Intrusion Detection Systems (IDS), although effective in identifying known attack signatures, often fail to detect new and evolving threats. To address these challenges, this project presents a Network Intrusion Detection System (NIDS) that integrates machine learning with synthetic data generation to provide an adaptive defense mechanism against cyberattacks.

The proposed system uses ensemble algorithms such as Random Forest and Gradient Boosting to classify network traffic as normal or malicious with high accuracy. Unlike conventional methods that rely only on existing datasets, the project introduces a synthetic traffic generation module to create diverse and balanced datasets for improved training. This enhances the system's ability to detect unseen attack patterns while reducing data imbalance and overfitting.

Additionally, the system includes graphical visualizations such as attack distributions, correlation heatmaps, and packet size plots, providing insights into traffic behavior and feature relationships. These visualizations improve interpretability and help administrators better understand intrusion patterns.

Through testing and evaluation, the proposed NIDS demonstrates improved accuracy, adaptability, and visualization compared to traditional IDS systems. Overall, the project contributes to cyber threat intelligence by combining data generation, machine learning, and visualization into a unified and scalable framework.

ACKNOWLEDGEMENTS

We would like to express our deepest gratitude to our guide, Ms. Geetanjali Tyagi for her valuable guidance, consistent encouragement, personal caring, timely help and providing us with an excellent atmosphere for doing research. All through the work, in spite of her busy schedule, she has extended cheerful and cordial support to us for completing this research work.

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TABLE OF CONTENTS

ABSTRACT iii

ACKNOWLEDGEMENTS iv

1 INTRODUCTION

1.1 Background 1

1.2 Introduction to Intrusion Detection Systems (IDS) 1

1.3 Problem Statement 1

1.4 Objective 2

1.5 Scope of the study 2

1.6 Proposed system overview 2

2 LITERATURE SURVEY

2.1 Review 3

2.2 Machine Learning-Based Anomaly Detection Approaches 3

2.3 Ensemble Learning Techniques in IDS 3

2.4 Challenges in IDS Data Availability 4

2.5 Summary of Research Gaps 4

3 **SYSTEM ANALYSIS** **5**

4 **SYSTEM DESIGN** **6**

5 **CODING & TESTING** **8**

6 **CONCLUSION** 11

7 **FUTURE ENHANCEMENT** **12**

8 **REFERNCES** **13**

CHAPTER 1: INTRODUCTION

1.1 Background of the Study

In today's digital era, the rapid expansion of interconnected systems and the widespread adoption of the internet have led to an exponential increase in data exchange across networks. From personal devices to large-scale enterprise systems, billions of devices continuously communicate, making network security a critical concern. Ensuring the confidentiality, integrity, and availability of data has become essential for maintaining trust in digital infrastructures.

With the rise of cloud computing, IoT devices, and distributed systems, the attack surface for cyber threats has significantly expanded. Cyberattacks have evolved in complexity and sophistication, often exploiting vulnerabilities in network protocols, software configurations, and human behavior. This dynamic threat landscape necessitates advanced and adaptive security mechanisms that go beyond traditional approaches.

1.2 Introduction to Intrusion Detection Systems (IDS)

An Intrusion Detection System (IDS) is a security solution designed to monitor network traffic and system activities for malicious behavior or policy violations. IDS can be broadly categorized into Network-based IDS (NIDS) and Host-based IDS (HIDS), depending on their deployment and monitoring scope.

Traditional IDS solutions primarily rely on signature-based detection techniques, where known attack patterns are stored in a database and matched against incoming traffic. While these systems are efficient in detecting known threats, they fail to identify new or evolving attack patterns, making them less effective in modern cybersecurity environments.

1.3 Problem Statement

Despite the advancements in cybersecurity, existing intrusion detection systems face several limitations. Signature-based systems lack adaptability and cannot detect zero-day or previously unknown attacks. On the other hand, anomaly-based systems, although capable of identifying unusual patterns, often suffer from high false positive rates.

Another critical issue is the lack of high-quality, diverse, and up-to-date datasets for training machine learning models. Many existing datasets do not accurately represent real-world network traffic, leading to poor generalization of detection models. These

challenges highlight the need for a more robust, intelligent, and data-driven intrusion detection approach.

1.4 Objectives of the Study

The primary objective of this project, titled **"Network Intrusion Detection System (NIDS)"**, is to develop an efficient and intelligent system capable of detecting network intrusions using machine learning techniques. The specific objectives include:

- To implement machine learning models such as Random Forest and Gradient Boosting for accurate classification of network traffic.
- To develop a synthetic data generation mechanism that simulates realistic network traffic for effective model training.
- To incorporate data visualization techniques for better understanding and interpretation of results.

1.5 Scope of the Study

This project focuses on the development of a Network Intrusion Detection System that leverages machine learning for anomaly detection. The system is designed to operate on generated and preprocessed network datasets, enabling the training and evaluation of multiple models.

The scope includes synthetic data generation, model training, performance evaluation, and visualization of results. However, the system is limited to a simulated environment and may require further enhancements for deployment in real-time large-scale networks.

1.6 Proposed System Overview

To address the limitations of traditional IDS, the proposed system integrates machine learning and data analytics into a unified platform. It utilizes synthetic data generation techniques to create realistic network traffic scenarios, which are then used to train models such as Random Forest and Gradient Boosting.

These models learn to distinguish between normal and malicious traffic patterns, enabling accurate classification of network activities. Additionally, the system includes visualization tools such as correlation heatmaps, attack distribution graphs, and packet size analysis, which help in interpreting model behavior and understanding data relationships.

The overall aim is to develop a system that is not only accurate and scalable but also interpretable, thereby enhancing its practical usability in applications.

CHAPTER 2: LITERATURE SURVEY

2.1 Review of Signature-Based Intrusion Detection Systems

Early intrusion detection systems (IDS), such as Snort (Roesch, 1999) and Bro IDS (now Zeek) (Paxson, 1999), primarily relied on signature-based detection techniques. These systems function by comparing network traffic against a database of known attack signatures, making them highly effective for detecting previously identified threats. Signature-based IDS are widely appreciated for their low false positive rates and efficiency in real-time environments. However, their major limitation lies in their inability to detect zero-day attacks or variations of known threats, as they require frequent updates to their signature databases. This rigidity makes them less adaptable in dynamic and evolving cybersecurity landscapes (Scarfone & Mell, 2007).

2.2 Machine Learning-Based Anomaly Detection Approaches

To address the shortcomings of signature-based systems, researchers introduced anomaly-based detection methods that leverage machine learning algorithms. Techniques such as Support Vector Machines (SVM), Decision Trees, and K-Nearest Neighbours (KNN) have been widely used for classifying network traffic as normal or malicious (Sommer & Paxson, 2010). These models are typically trained on benchmark datasets like KDD Cup 1999 (KDD99) and its improved version NSL-KDD (Tavallaee et al., 2009).

Anomaly-based systems offer the advantage of detecting previously unseen attacks by identifying deviations from normal behavior. However, they suffer from challenges such as high false positive rates, data imbalance, and limited representation of modern attack scenarios in traditional datasets. Studies have shown that the outdated nature of datasets like KDD99 can negatively impact the generalization of machine learning models in real-world environments (Ring et al., 2019).

2.3 Ensemble Learning Techniques in IDS

Recent advancements in IDS research have focused on ensemble learning methods, which combine multiple machine learning models to improve detection performance. Techniques such as Random Forest, Gradient Boosting, and Extreme Gradient Boosting (XGBoost) have demonstrated superior accuracy and robustness compared to single classifiers (Chen & Guestrin, 2016).

Ensemble models reduce variance and bias by aggregating predictions from multiple learners, leading to improved detection rates and lower false positives. For instance, Zhang et al. (2019) demonstrated that Random Forest-based IDS achieved higher precision and recall compared to standalone classifiers. Similarly, Kumar et al. (2022) showed that hybrid ensemble approaches significantly enhance intrusion detection performance in complex network environments. These findings highlight the growing importance of ensemble techniques in modern IDS design.

2.4 Challenges in IDS Data Availability and Synthetic Data Generation

One of the major challenges in intrusion detection research is the lack of realistic, diverse, and up-to-date datasets. Many widely used datasets do not accurately reflect current network behaviors and attack patterns, limiting the effectiveness of trained models. To overcome this issue, researchers have explored synthetic data generation techniques using simulation tools, generative models, and traffic emulation systems (Ferrag et al., 2020).

Synthetic datasets help address issues such as class imbalance and insufficient attack representation, enabling better model training and evaluation. In addition, data visualization techniques—such as correlation heatmaps, feature importance plots, and traffic distribution graphs—play a crucial role in understanding relationships between network attributes and improving model interpretability (Han et al., 2011). These techniques allow researchers to gain deeper insights into feature behavior and model decision-making processes.

2.5 Summary of Research Gaps

Despite significant advancements in IDS technologies, several challenges remain unresolved. Many existing systems focus primarily on improving detection accuracy while neglecting issues such as dataset quality, adaptability to evolving threats, and interpretability of results. Additionally, there is a lack of integrated platforms that combine data generation, model training, and performance visualization in a unified manner. This fragmentation limits the practical applicability and usability of IDS solutions in real-world scenarios.

Section	Subtopic	Summary
2.1	Signature-Based IDS	Detects known attacks using predefined signatures but fails against new threats.
2.2	Machine Learning-Based Anomaly Detection	Uses ML models like SVM, KNN, and Decision Trees to identify unknown attacks, but suffers from data imbalance and false positives.
2.3	Ensemble Learning Techniques	Combines multiple models (Random Forest, XGBoost) to improve detection accuracy and reduce false positives.
2.4	Synthetic Data Generation & Visualization	Generates realistic training data and uses visualization tools to enhance model understanding and performance analysis.
2.5	Research Gaps	Highlights limitations in existing IDS such as lack of integration, poor data quality, and adaptability issues.

CHAPTER 3: SYSTEM ANALYSIS

Problem Definition
Traditional IDSs struggle with dynamic and evolving threats, limited training data, and lack of interpretability. The proposed system solves these issues by generating synthetic datasets, using robust ML algorithms, and producing intuitive visual analyses.

Objectives
- Generate synthetic network traffic data representing normal and malicious activity.
- Train Random Forest and Gradient Boosting models for intrusion classification.
- Provide visual insights such as attack type distribution and feature correlations.
- Offer a modular and easily extendable architecture.

Existing System
Existing systems rely heavily on static datasets and fixed detection rules, which are often outdated and limited to known attacks. They lack adaptability and data visualization for deeper insights.

Proposed System
The proposed system automatically generates synthetic traffic, processes the data, trains models, and visualizes outcomes. It ensures real-time analysis, improved accuracy, and interpretability through graphical outputs.

Functional Requirements
- Dataset generation and labeling
- Model training, testing, and evaluation
- Detection of intrusion types
- Visualization of performance metrics

Non-Functional Requirements
- High detection accuracy
- Low processing time
- Scalability for large datasets
- Usability through intuitive outputs

CHAPTER 4: SYSTEM DESIGN

The **Network Intrusion Detection System** is designed with modularity and scalability in mind, ensuring that each part functions independently yet contributes to an integrated detection workflow.

System Architecture

1. **Data Generation Module** – Creates synthetic network traffic simulating normal and various attack behaviors.
2. **Feature Engineering Module** – Extracts attributes such as packet size, protocol type, source/destination IPs, and flow duration.
3. **Model Training Module** – Trains machine learning classifiers using Random Forest and Gradient Boosting algorithms.
4. **Detection & Evaluation Module** – Applies trained models to classify unseen data or live traffic.
5. **Visualization Module** – Generates graphs such as attack type frequency, correlation heatmaps, and packet size distributions for analytical insight.

Data Flow Representation

Synthetic Data → Preprocessing & Feature Extraction → ML Model Training → Detection → Visualization & Analysis

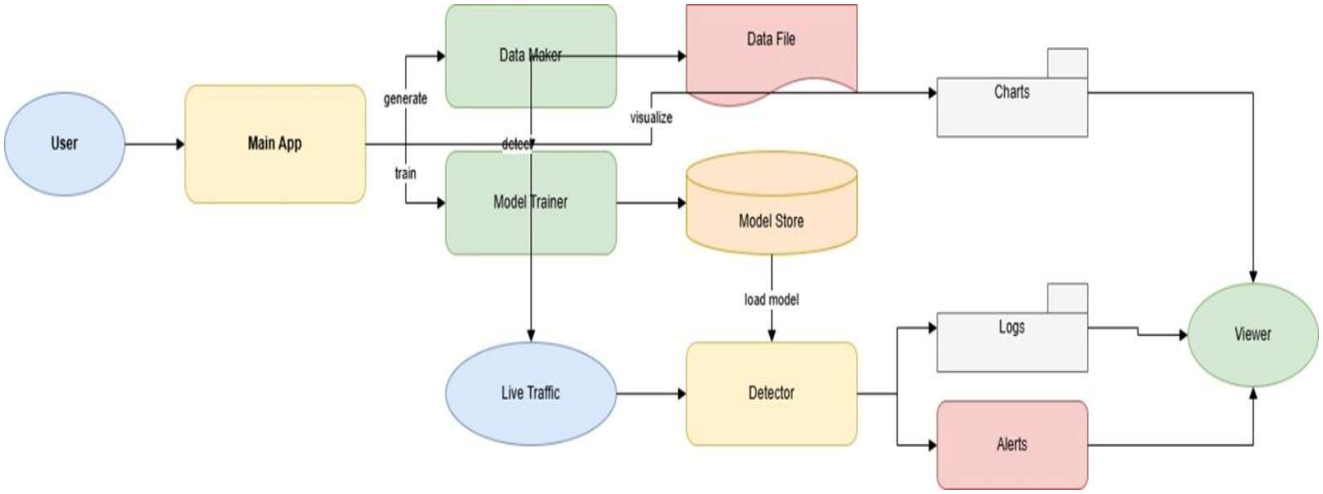
Design Considerations

- Ensure efficient memory usage and scalability for large data volumes.
- Maintain high interpretability through labeled visual outputs.
- Modularize code to allow easy integration of new algorithms or attack types.

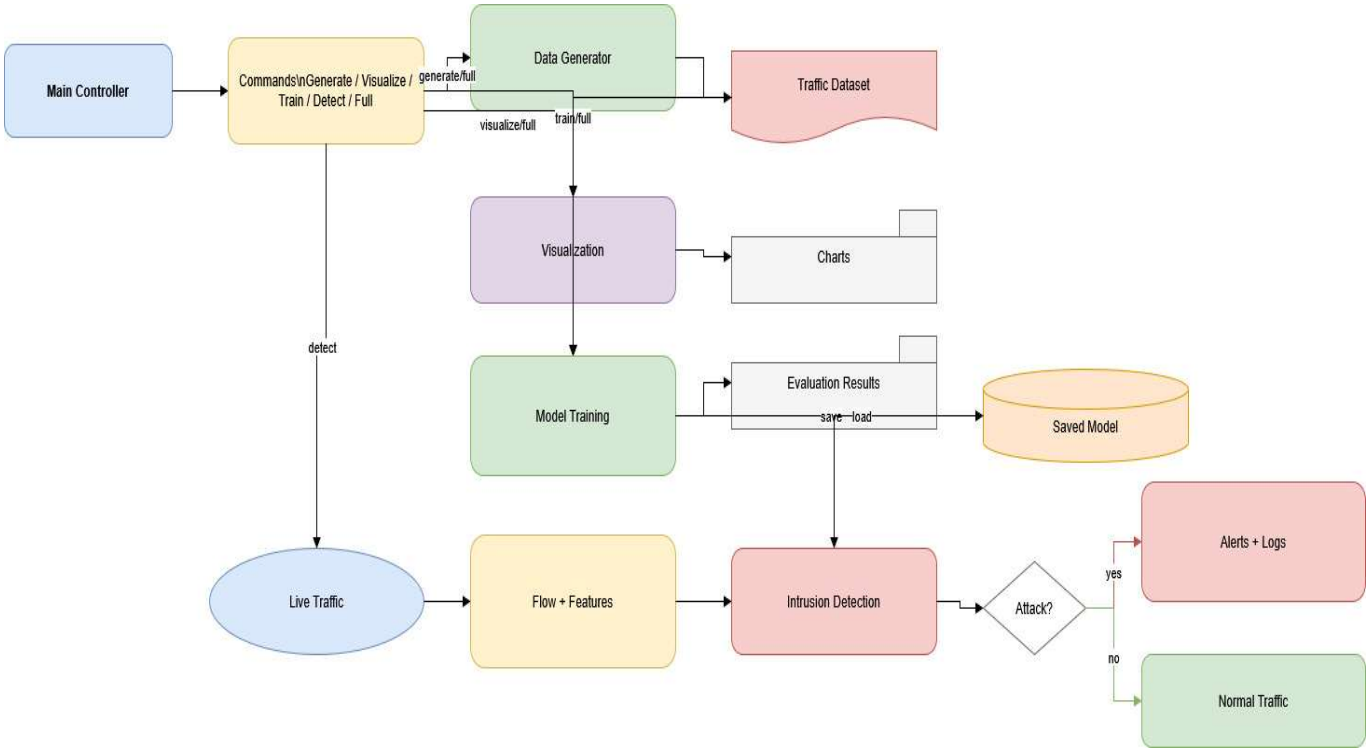
Tools and Technologies

- **Programming Language:** Python
- **Libraries:** Pandas, NumPy, Scikit-learn, Seaborn, Matplotlib
- **Environment:** Jupyter Notebook / VS Code
- **Storage:** CSV/SQLite database for dataset management

ARCHITECTURE DIAGRAM



BLOCK DIAGRAM



CHAPTER 5: CODING AND TESTING

Coding Implementation

The system was implemented in **Python**, leveraging open-source libraries for ML and visualization. Key components include:

- data_generator.py — Generates synthetic data with normal and malicious traffic.
- train_model.py — Trains models (Random Forest, Gradient Boosting) and evaluates accuracy.
- detect_intrusion.py — Detects intrusion on new data samples or live packets.
- visualization.py — Produces detailed plots like attack distribution and feature correlation.
- main.py — Acts as the control script linking all modules.

Model Workflow:

1. Generate synthetic data using adjustable parameters.
2. Preprocess data by encoding features and normalizing values.
3. Train models using 70–30 train-test split.
4. Evaluate accuracy, precision, recall, and F1-score.
5. Visualize the model's behavior through analytical plots.

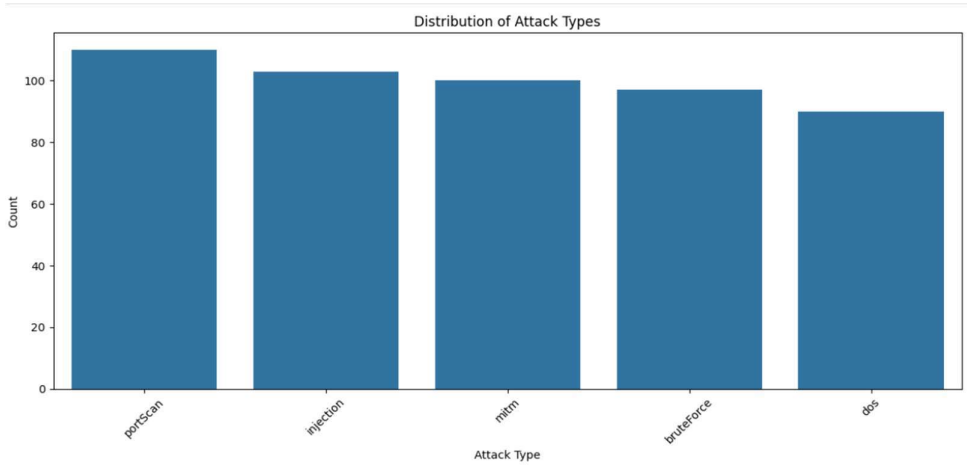
Testing and Validation

The testing phase validated both performance and reliability:

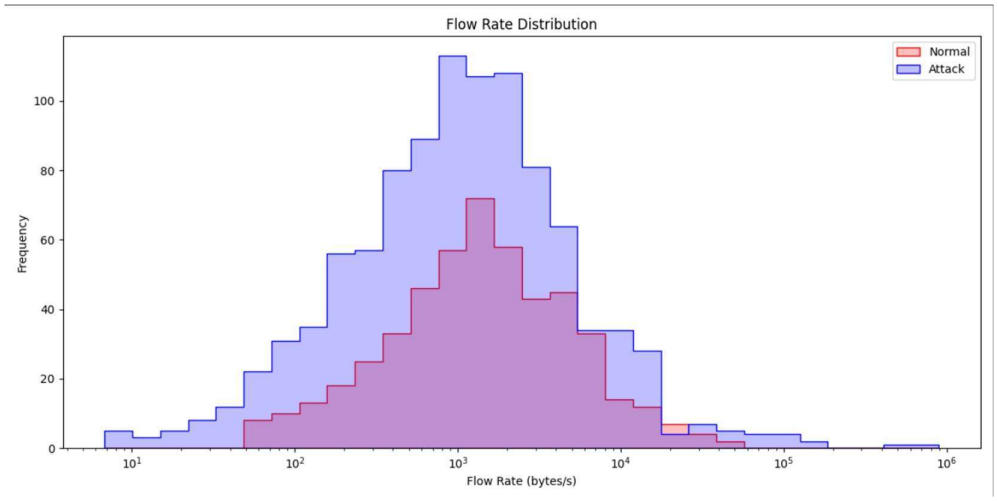
- **Unit Testing:** Verified each module's independent functionality.
- **Integration Testing:** Ensured smooth data flow between modules.
- **Performance Testing:** Evaluated model response time and computational efficiency.
- **Accuracy Testing:** Compared model outputs against known labels.
- **Visualization Validation:** Ensured graphs accurately represented data relationships.

Both **Random Forest** and **Gradient Boosting** demonstrated high accuracy with balanced precision and recall, confirming that ensemble approaches enhance intrusion detection effectiveness.

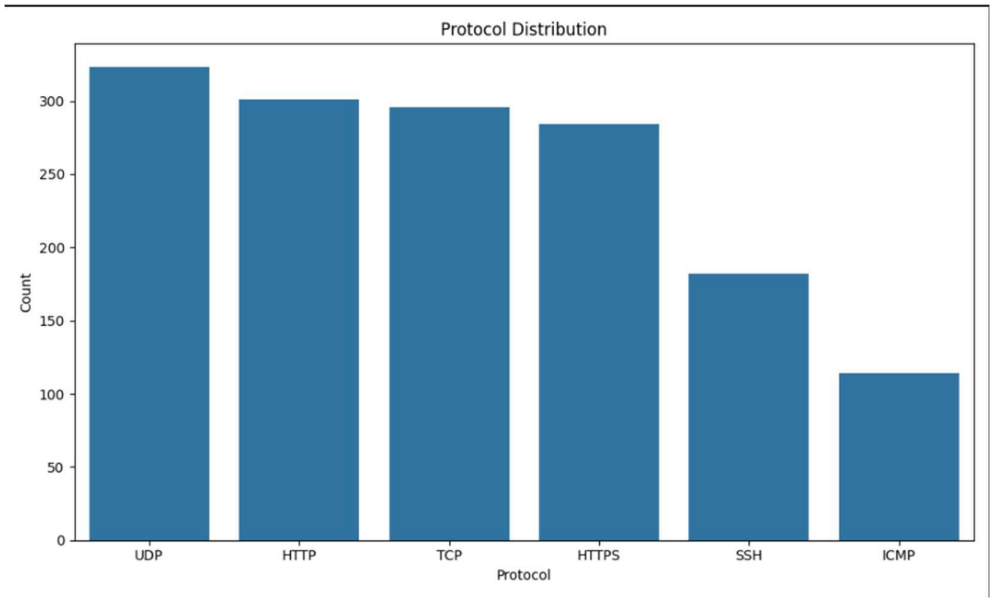
GRAPH FOR DISTRIBUTION OF ATTACK TYPES



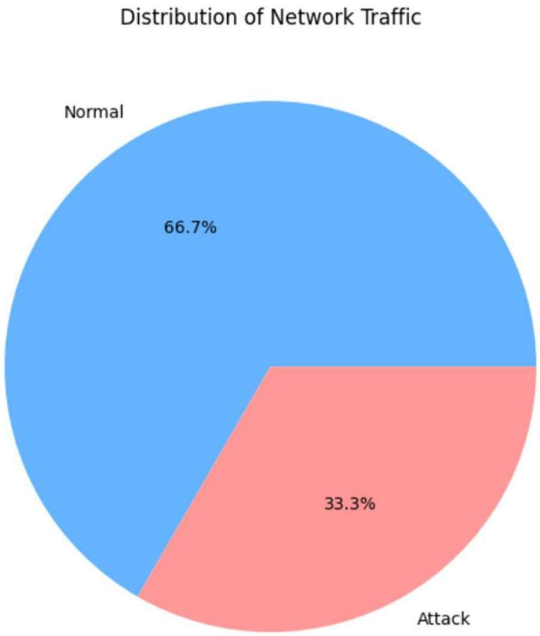
GRAPH FOR FLOW RATE DISTRIBUTION



GRAPH FOR PROTOCOL DISTRIBUTION



PIE CHART FOR DISTRIBUTION OF NETWORK TRAFFIC



CHAPTER 6: CONCLUSION

The **Network Intrusion Detection System (NIDS)** developed in this project represents a modern and proactive approach to securing digital networks against a constantly evolving range of cyber threats. By integrating **machine learning algorithms** such as Random Forest and Gradient Boosting with **synthetic network traffic generation**, the system effectively addresses key challenges in intrusion detection—particularly the detection of zero-day attacks and data scarcity for model training.

One of the system's most valuable contributions lies in its **ability to generate synthetic network traffic**, which enhances dataset diversity and ensures that the model is trained on a broad spectrum of attack and normal patterns. This significantly improves detection accuracy and system reliability in real-world network environments. Furthermore, the inclusion of **graphical analytics**—such as attack-type distributions, correlation heatmaps, and packet-size visualizations—provides network administrators with meaningful insights into system behavior and attack characteristics, transforming raw data into actionable intelligence.

The results obtained from testing indicate that the proposed IDS performs efficiently, balancing speed, precision, and interpretability. It not only detects malicious activity effectively but also assists in understanding the nature of attacks, contributing to smarter security decision-making. The system's modular design allows for easy integration into existing network infrastructures and provides a solid foundation for continuous learning and enhancement.

In conclusion, this project successfully demonstrates the potential of **machine learning-driven IDS frameworks** in improving cybersecurity resilience. With further refinements such as deep learning integration, real-time threat prediction, and adaptive feedback mechanisms, the system can evolve into a **fully autonomous and intelligent intrusion detection platform**, paving the way for safer and smarter digital networks in the future

CHAPTER 7: FUTURE ENHANCEMENT

- **Deep Learning Integration:**
Implement deep neural networks (CNN, LSTM) for advanced temporal pattern detection in traffic sequences.

- **GAN-Based Data Generation:**
Use Generative Adversarial Networks to create realistic synthetic network flows for better model generalization.

- **Real-Time Monitoring:**
Integrate live packet capture tools (Wireshark, tcpdump) for real-time attack detection.

- **Cloud-Based Deployment:**
Host the IDS as a cloud service for distributed and large-scale monitoring.

- **Adaptive Learning:**
Enable automatic retraining when new attack types are detected, keeping models up to date.

- **User Dashboard:**
Build an interactive web-based dashboard to visualize detections and metrics dynamically.

- **Hybrid Detection Approach:**
Combine rule-based (signature) and machine learning (anomaly) detection for higher reliability.

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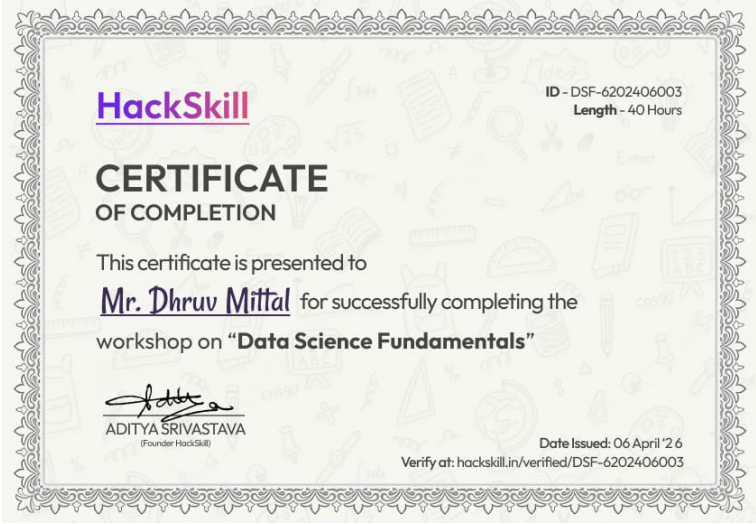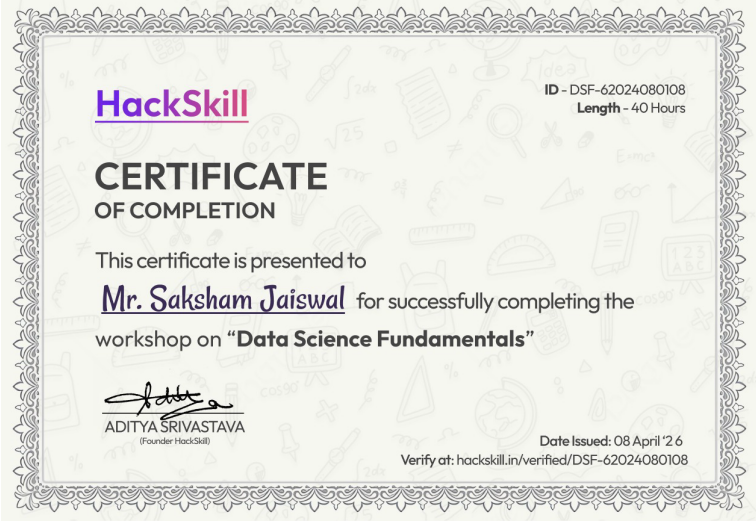
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